



Project Proposal

Predicting Learner Support Needs in CNIB's Assistive Technology Academy

Introduction

The **Canadian National Institute for the Blind (CNIB)** is a national, community-based charity founded in 1918. CNIB works to change what it is to be blind today by delivering innovative programs and powerful advocacy that empower people who are blind, Deafblind, or have low vision to live the lives they choose. CNIB provides services and supports that enhance independence, literacy, employment, and community participation, and works alongside Canadians with sight loss, allies, advocates, and partners from coast to coast to coast. Through programs such as the Assistive Technology Academy and the Come to Work employment program, CNIB focuses on removing barriers and building the skills people with sight loss need to participate fully in today's workforce.

What is the problem?

CNIB's Assistive Technology Academy is a national, cohort-based training program designed to equip people who are blind, Deafblind, or have low vision with essential digital skills for employment and independence. The Academy was created to address a persistent barrier to employment: while assistive technology is critical to participation in today's workforce, access to structured, high-quality training remains inconsistent and slow.

Although the Academy's intensive six-week "bootcamp" model has proven effective, some participants fall behind in skill development or struggle to complete homework, workbooks, and practice tasks. These challenges do not necessarily reflect a lack of effort or ability. They often relate to differences in learning pace, learning style,



confidence, or difficulty mastering specific concepts such as navigation, screen reader commands, or task sequencing.

Currently, trainers rely on observation, conversations, and manual reviews to understand who is struggling and why. In a fast-paced, cohort-based program, this makes it difficult to consistently detect patterns such as repeated mistakes, inconsistent practice behaviours, differences in progress between learners who complete homework and those who do not, and recurring points of frustration or confusion.

As a result, opportunities for timely, targeted intervention can be missed. **To address this gap, CNIB needs a way to consistently identify early warning signs of learning challenges, understand the underlying barriers learners face, and surface practical solutions that inform timely adjustments to training and support.**

Why is this important?

Assistive technology skills are essential for employment, independence, and full participation in today's workforce for people who are blind, Deafblind, or have low vision. For people with visual disabilities, technology is not optional; it is the primary way to access information, communicate at work, and perform job tasks.

Employment outcomes for people with visual disabilities remain significantly lower than those of the general population. In Canada, only about 54% of working-age adults with a seeing disability are employed, and the unemployment rate for Canadians with sight loss is approximately three times the national average. These gaps highlight the ongoing barriers to workforce inclusion faced by this community.

CNIB's Assistive Technology Academy was designed to reduce these barriers by accelerating access to workplace-ready digital skills. However, in a compressed six-week training model, even short delays in identifying learning challenges can affect participant confidence, engagement, and program completion. Learners who fall behind without timely support may disengage despite having the potential to succeed.



Strengthening the ability to identify learning challenges early and respond with targeted, personalized support helps ensure participants stay engaged and complete the program successfully. This approach enables learners to leave with practical, employer-valued digital skills and directly supports improved employment outcomes for people with visual disabilities.

Why can machine learning help?

In a fast-paced, cohort-based program, it is difficult to consistently identify early signs of struggle across multiple learners solely through manual observation. Machine learning can analyze multiple indicators at once, such as practice frequency, homework completion, repeated errors, and navigation difficulties, and identify patterns over time that may not be immediately visible.

Ultimately, this analysis will help CNIB identify the key challenges preventing some learners from keeping up and surface solutions that can shape training approaches, support strategies, and curriculum adjustments, ensuring that all participants, regardless of learning style or pace, have a fair opportunity to succeed.

What data can be used?

This project would use internal program data generated through the Assistive Technology Academy that reflects learning activity and progress rather than personal identity. Examples include:

- Homework and workbook completion records
- Practice frequency and consistency
- Patterns of common or repeated errors
- Structured insights drawn from training session notes or learning interactions

While the AI would evaluate learning activity at the individual level to identify patterns and risks, it would not be designed to process personal or sensitive information. The focus is on activities, behaviours, and learning outcomes. All data would be handled in accordance with CNIB's privacy and data governance standards